1.65 Constant Acceleration

Worked Solutions

1 a)

Acceleration is gradient: $a = \frac{v-u}{t}$

$\Rightarrow at = v - u \Rightarrow v = u + at$ as req.

1 b)

Displacement is area under curve:

Trapezium: $s = \frac{1}{2}(u + v)t$ as req.

1 c)

Displacement is area under curve:

Rectangle + Triangle: $s = ut + \frac{1}{2}t \times (v - u)$

But $at = v - u$

so $s = ut + \frac{1}{2}t \times at = ut + \frac{1}{2}at^2$ as req.

1 d)

Displacement is area under curve:

Rectangle − Triangle: $s = vt - \frac{1}{2}t \times (v - u)$

But $at = v - u$

so $s = vt - \frac{1}{2}t \times at = vt - \frac{1}{2}at^2$ as req.

2)

$at = v - u$ and $t = \frac{v-u}{a}$

$s = v\left(\frac{v-u}{a}\right) - \frac{1}{2}(v - u)\left(\frac{v-u}{a}\right)$

$\Rightarrow as = v(v - u) - \frac{1}{2}(v - u)(v - u)$

$\Rightarrow as = v^2 - uv - \frac{1}{2}v^2 + uv - \frac{1}{2}u^2$

$\Rightarrow as = \frac{1}{2}v^2 - \frac{1}{2}u^2 \Rightarrow 2as = v^2 - u^2$

$\Rightarrow v^2 = u^2 + 2as$

3 a)

$6 = 4 + 2t \Rightarrow t = 1$

3 b)

$5 = 3u - 9 \Rightarrow u = \frac{14}{3}$

3 c)

$12 = 4 \times 5 - 8a \Rightarrow a = 1$

3 d)

$9 = 4 + 3s \Rightarrow s = \frac{5}{3}$

3 e)

$20 = 4(5 + v) \Rightarrow v = 0$

4)

$20 = 15 - 18a \Rightarrow a = -\frac{5}{18}$

$20 = 3(u + 2.5) \Rightarrow u = \frac{25}{6}$

5)

$15 = \frac{1}{2}(1.5 + 4.5)t \Rightarrow t = 5$

$20.25 = 2.25 + 30a \Rightarrow a = \frac{3}{5}$

6 a)

$v = 6 + 2 \times 7 = 20 \text{ ms}^{-1}$

6 b)

$s = 6 \times 7 + \frac{1}{2} \times 2 \times 7^2 = 91 \text{ m}$

7)

$100 = 1.1t + 0.025t^2 \Rightarrow t = -89.0 \text{ or } t = 45.0$

$t \geq 0$ so 45.0 s (3 SF)

8 a)

$s = ut + \frac{1}{2}at^2 = 5 \times 10 + \frac{1}{2} \times 0.1 \times 10^2 = 55 \text{ m}$

8 b)

$v = u + at = 5 + 0.1 \times 10 = 6 \text{ ms}^{-1}$

9 a)

$4 = 0 + 3a \Rightarrow a = \frac{4}{3} \text{ ms}^{-2}$

$s = \frac{3}{2}(0 + 4) \Rightarrow s = 6 \text{ m}$

9 b)

$0.014 = 0 + 2a \Rightarrow a = 0.007 \text{ ms}^{-2}$

$s = \frac{2}{2}(0 + 0.014) \Rightarrow s = 0.014 \text{ m}$

9 c)

$10 = 0 + at \Rightarrow a = \frac{10}{t} \text{ ms}^{-2}$

$s = \frac{t}{2}(0 + 10) \Rightarrow s = 5t \text{ m}$

10)

$3u = u + 12a \Rightarrow a = \frac{u}{6}$ ms^{-2}

11 a)

$s = 38, v = 0, u = \frac{200}{9}$

$38 = \frac{1}{2}\left(\frac{200}{9} + 0\right)t \Rightarrow t = \frac{171}{50} = 3.42$ s

11 b)

Stopping times are affected by the type of vehicle, the condition of the brakes, the weather and condition of the road. Also the driver needs 'thinking time' to apply the brakes.

12)

$s = 2000, u = \frac{70}{9}, v = \frac{80}{9}$

$\left(\frac{80}{9}\right)^2 = \left(\frac{70}{9}\right)^2 + 4000a \Rightarrow a = \frac{1}{216} = 0.00463$ ms^{-2} (3SF)

13 a)

B to C: $16 = 12 + 15a \Rightarrow a = \frac{4}{15} = 0.267$ ms^{-2} (3 SF)

13 b)

A to B: $12^2 = u^2 + 2 \times \frac{4}{15} \times 150 \Rightarrow u = 8$ ms^{-1}

14 a)

$10.4 = 5 \times 4 + \frac{1}{2} \times 4^2 a \Rightarrow a = -1.2$ ms^{-2}

14 b)

$5 = u - 1.2 \times 6 \Rightarrow u = 12.2$ ms^{-1}

14 c)

B to C: $10.4 = \frac{1}{2}(5 + v) \times 4 \Rightarrow v = 0.2$ ms^{-1}

$v^2 = u^2 + 2as \Rightarrow 0 = 0.2^2 - 2 \times 1.2s \Rightarrow s = \frac{1}{60}$ m

0.0167 m (3 SF) beyond C

15 a)

A to B: $30 = 15 + at$

B to C: $50 = 30 + at$

$\frac{15}{a} : \frac{20}{a}$ so 3 : 4

15 b)

A to C: $50^2 = 15^2 + 2 \times 150a \Rightarrow a = \frac{91}{12}$

A to B: $30^2 = 15^2 + 2 \times \frac{91}{12}s \Rightarrow s = \frac{4050}{91}$

44.5 m (3SF)

16 a)

Final speed of part one = Initial speed of part two

Part one: $25 = 2v - 2a$

Part two: $110 = 5u + \frac{25}{2}a$

Solving simultaneously: $u = \frac{213}{14}$, $a = \frac{19}{7}$

So $a = 2.71$ ms^{-2}

16 b)

$25 = 2u + 2 \times \frac{19}{7} \Rightarrow u = \frac{137}{14}$

so initial speed is 9.79 ms^{-1} (3SF)

17)

CAR A: $s = ut$

CAR B: $s = -15 + (u - 0.2)t + \frac{1}{2} \times 0.8t^2$

Solving simultaneously:

$ut = -15 + (u - 0.2)t + \frac{1}{2} \times 0.8t^2$

$0 = -15 - 0.2t + 0.4t^2$

$t = -5.88$ or $t = 6.38$ so 6.38 seconds

18 a)

First phase: $48 = 6k + \frac{1}{2}ak^2 \Rightarrow a = \frac{96-12k}{k^2}$

18 b)

$48 = \frac{1}{2}(v + 6)k \Rightarrow u = \frac{96}{k} - 6$

18 c)

$204 = \left(\frac{96}{k} - 6\right) \times 10 + \frac{1}{2}\left(\frac{96-12k}{k^2}\right) \times 10^2$

$204k^2 = (96k - 6k^2) \times 10 + 50(96 - 12k)$

$204k^2 = 960k - 60k^2 + 4800 - 600k$

$264k^2 - 360k - 4800 = 0 \Rightarrow k = 5$ or $k = -\frac{40}{11}$

But only the positive k gives a result that fits the model. So $k = 5$

18 d)

$a = \frac{96-12\times5}{25} = 1.44$ ms^{-2}

19)

$3 = 0 + \frac{1}{2}a \times 2.2^2$ so $a = \frac{150}{121} = 1.24$ ms^{-2} (3 SF)

20 a)

$s = 0 + \frac{1}{2} \times -9.81 \times 0.8^2 = -3.1392$ m

Height is 3.1392 m

3.14 m (3 SF – AQA)

20 b)

$v = -9.81t$

Speed $= 9.81t$

20 c)

$v = -9.81 \times 0.8 = -7.848$ ms^{-1}

Speed is 7.848 ms^{-1}

7.85 ms^{-1} (3 SF – AQA)

21 a)

$v^2 = 0 + 2 \times 9.8 \times 16 \Rightarrow v = \frac{28}{5}\sqrt{5}$

So $v = 12.5$ ms^{-1} (3 SF)

$v = 13$ ms^{-1} (2 SF – AQA)

21 b)

$8 = 0 + \frac{1}{2} \times 9.8t^2 \Rightarrow t = \frac{4}{7}\sqrt{5}$

So $t = 1.28$ s (3 SF)

$t = 1.3$ s (2 SF – AQA)

21 c)

A more accurate value of the acceleration of gravity (including considering where on the planet the object is dropped as g is variable).

Calculating for the object having a shape rather than being modelled as a particle.

22 a)

Reaches ground when $s = 0$

$0 = 14.7t + \frac{1}{2} \times -9.8t^2 \Rightarrow t = 0, t = 3$

So $t = 3$ s

22 b)

It travels to maximum height:

$v^2 = u^2 + 2as$ gives $0 = 14.7^2 + 2 \times -9.8s$

$\therefore s = 11.025$ m

$s = ut + \frac{1}{2}at^2$ gives $11.025 = 14.7 \times t - \frac{1}{2} \times 9.8t^2$

$\therefore t = 1.5$ seconds to reach maximum height

Final 0.5 seconds:

$s = 0 + \frac{1}{2} \times 9.8 \times 0.5^2 = 1.225$ m

So total distance is 12.25 metres

23 a)

Max height when $v = 0$

$0 = 24.5 - 9.8t \Rightarrow t = 2.5$ s

23 b)

$s = 24.5 \times 6.5 + \frac{1}{2} \times -9.8 \times 6.5^2 = -47.775$

So height of bridge is $s = 47.8$ m (3 SF)

$s = 48$ m (2 SF – AQA)

23 c)

Distance to max height: $0 = 24.5^2 + 2 \times -9.8s$

$\Rightarrow s = 30.625$

Total distance: $2 \times 30.625 + 47.775 = 109.025$ m

So distance is 109 m or 110 m (2 SF – AQA)

24 a)

Hitting the ground: $s = -1.65$

$-1.65 = 6.25t + \frac{1}{2} \times -9.8t^2 \Rightarrow t = 1.5$ or $t = -\frac{11}{49}$

So $t = 1.5$ seconds after being thrown.

24 b)

0.35 m above the start point.

$0.35 = 6.25t + \frac{1}{2} \times -9.8t^2$

$\Rightarrow t = 1.2168\ldots$ s, $t = 0.05870\ldots$ s

So $t = 1.22$ s, $t = 0.0587$ s (3 SF)

$t = 1.2$ s, $t = 0.059$ s (2 SF – AQA)

24 c)

$1.2168\ldots - 0.05870\ldots = 1.15$ seconds (3 SF)

$t = 1.2$ seconds (2 SF – AQA)

25 a)

$22^2 = u^2 + 2 \times -10 \times -15 \Rightarrow u = \pm 2\sqrt{46}$

Speed is 13.6 ms^{-1} (3 SF)

10 ms^{-1} (1 SF – AQA)

25 b)

Greatest height when $v = 0$

$0 = (2\sqrt{46})^2 + 2 \times -10 \times s \Rightarrow s = 9.2$

So 24.2m (3 SF) above the ground

20 m (1 SF – AQA)

25 c)

$-15 = -22t + \frac{1}{2} \times 10t^2 \Rightarrow t = 3.5565, t = 0.8435$

$t = 3.56$ seconds

4 s (1 SF – AQA)

25 d)

Consider reducing the acceleration due to gravity (as resistance would change the total acceleration)

The initial speed would be larger to give the same final speed, this would lead to a higher greatest height reached, and it would take longer to hit the ground.

26 a)

$v = 0 - 10 \times 4 = -40$ so speed is 40 ms^{-1}

26 b)

Phase 1:

$s = 0 \times 4 + \frac{1}{2} \times (-10) \times 4^2 = -80$ so 80 m

Phase 2:

$u = -40, a = 12, t = 3$

$s = -40 \times 3 + \frac{1}{2} \times 12 \times 3^2 = -66$ so 66 m

$v = u + at = -40 + 12 \times 3 = -4$ so 4 ms^{-1}

Phase 3:

Distance left: $2800 - 80 - 66 = 2654$ m

$-2654 = -4t$ so $t = 663.5$

Total time: $4 + 3 + 663.5 = 670.5$ seconds

700 s (1 SF – AQA)

[Please note that the numbers used here are unrealistic due to the impact of air resistance when skydiving]

27 a)

Ball A: $s = 20t - 5t^2$

Ball B: $s - 10 = -5t - 5t^2$

Solving simultaneously,

$20t - 5t^2 = -5t - 5t^2 + 10$

$25t = 10$ so $t = 0.4$ seconds.

27 b)

$s = 20 \times 0.4 - 5 \times 0.4^2 = 7.2$ m

$s = 7$ m (1 SF – AQA)

28 a)

Ball A: $s = 20t - 5t^2$

Ball B: $s - 5 = 12t - 5t^2$

When $s = 5$

Ball A: $0 = 5t^2 - 20t + 5$ so $t = 2 \pm \sqrt{3}$

So $t = 0.2679\ldots$, $t = 3.73\ldots$

Ball B: $0 = 12t - 5t^2$ so $t = 0, t = 2.4$

So they are both above the height for $2.4 - 0.2679$ which is $t = 2.13$ seconds

2 seconds (1 SF – AQA)

28 b)

Solving simultaneously,

$20t - 5t^2 = 12t - 5t^2 + 5$

$8t = 5$ so $t = \frac{5}{8} = 0.625$ seconds

0.6 seconds (1 SF – AQA)

28 c)

When $t = \frac{5}{8}$, $s = 20 \times \frac{5}{8} - 5 \times \left(\frac{5}{8}\right)^2 = \frac{675}{64}$

$s = 10.5$ metres from the ground

$s = 10$ m (1 SF – AQA)

29)

$v^2 = 0 + 2 \times -g \times -h \Rightarrow v = \sqrt{2gh}$

30)

$8^2 = 0^2 - 2 \times -g \times h \Rightarrow h = \frac{64}{2g} = \frac{32}{g}$ as req.

31 a)

$0 = u^2 - 2gs \Rightarrow s = \frac{u^2}{2g}$

31 b)

$0 = ut - \frac{1}{2}gt^2 \Rightarrow t = \frac{-u \pm \sqrt{u^2 - 0}}{-g} \Rightarrow t = \frac{u \mp \sqrt{u^2}}{g}$

So returned when $t = \frac{2u}{g}$

32)

Particle A: $-k = 0 - \frac{1}{2}gt^2$

Particle B: $-\frac{k}{10} = 0 - \frac{1}{2}g(t - 0.5)^2$

$-\frac{\frac{1}{2}gt^2}{10} = -\frac{1}{2}g(t - 0.5)^2$

$t^2 = 10(t - 0.5)^2$

$t^2 = 10t^2 - 10t + 2.5$

$0 = 9t^2 - 10t + 2.5$ so $t = \frac{10 \pm \sqrt{10}}{18}$

So $t = 0.7312376\ldots$, $t = 0.3798734633$

$t = 0.3798734633$ is before particle B is released.

When $k = 0.2673542488g$

So $k \approx 0.267g$ as req.